

Workbook answers

Unit 1 Environments and habitats

Topic 1.1 Habitats

Focus

- 1 Pictures drawn of any from the following: plants, trees, bushes, insects, earthworm, butterfly, frog, rabbit, bird, squirrel.

Practice

- 2 a A habitat is the natural home of an animal or **plant**.
 b The desert habitat is hot and **dry**.
 c The forest habitat is **cool**.
 d The desert habitat has fewer **living things**.
 e There are many living things in the **forest**.
 f The forest animals have many plants to **eat**.

Challenge

3

Living thing	What each living thing needs	Where they find it
bird	food	on the ground and around the tree
	water	pond or rain
	home	in the tree
tree	light	from the Sun
	water	from the soil
	soil	underground
rabbit	food	plants
	water	pond or rain
	home	burrow/ underground

Topic 1.2 Plants in different habitats

Focus

- 1 Light, water, soil and air drawn and/or labelled around the plant.
 2 Words circled are soil, Sun and water.

Practice

- 3 Drawing might include plants and gardening equipment, such as string and canes, but the three key things that must be present are soil, water and light.

Challenge

- 4 a In the hot desert the seed will **die**.
 b In the cold habitat the seed will **die**.
 c By a river the seed will **grow**.
 d The seed by the river grows best because it has **water**.

Topic 1.3 Animals in different habitats

Focus

- 1 Lines link duck to river; crab to seashore; bird to tree.

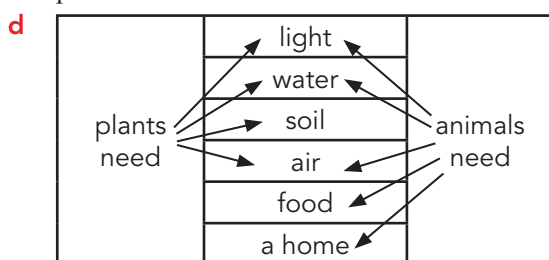
Practice

- 2 a The habitat for the mice is by the tree, in the **grass**.
 b The habitat for the frog is in and by the **river**.
 c The habitat for the rabbits is around the trees and across the **fields**.
 d The bees have a nest in the tree but fly all around looking for **flowers**.
 e The eagle hunts around many fields, rivers and **forests**.
 f The animal with the largest habitat is the **eagle**.

Challenge

- 3 a Lots of plants live by the **river**. These plants have a good habitat so they can live and **grow**.
 b Rabbits and mice eat seeds and **plants**. The snake and eagle eat small **animals**.

- c** A living thing finds what it needs in the right **habitat**. Without the things it needs a plant or animal will **die**.



Topic 1.4 Rocks and the environment

Focus

1

How we use the rock	Rock
	Chalk A very soft rock used to make chalk sticks for writing.
	Marble A hard, strong rock used in floor tiles, wall tiles and to make statues.
	Coal A black rock which we can burn to keep us warm
	Slate Can be blue, purple or grey. It is smooth to touch and often used in floor and roof tiles.
	Granite A very strong rock used to make buildings. It has bits in it that are easy to see.

Practice

- 2 a** This is a quarry.
On the surface workers use diggers and explosives to break up the **rocks**.
Limestone is a rock we dig up in quarries.

- b** This is a mine.
Underground workers break the rock with explosives, tools and **machines**.
Coal is a rock we dig up in mines.
- c** This is a riverbed where workers dig out sand, rocks and **stones**.
Sand, rocks and stones are used for building.

Challenge

- 3 a** The habitat of plants and animals will be **lost**.
b The river will be **polluted**.
c Many animals may have to move to look for a new **home**.
d Very few animals and plants will be able to stay, many will **die**.
e Animals and plants may never live here **again**.

Topic 1.5 Can we care for our environment?

Focus

- 1** All speech bubbles should be ticked, except for 'There is nothing we can do.' should be ticked.
2 Any three reasonable ideas drawn, such as plant trees, feed birds, tidy litter, stop cutting down trees.

Practice

3



recycle plastic



recycle paper

- 4 a** Tuesday
b Friday

- c Because on these days the class did not use much paper / the children were out of class for part of the day.
- d Any reasonable suggestion, for example, use both side of paper, use computer more without printing.

Challenge

- 5 a The number of animals is going down.
b Any number below 28
c Some may be dead. Others will have moved.
d Accept any reference to loss of habitat, nowhere to live or no food to eat.
e Any reasonable ideas, such as find a new home for the animals, stop cutting down the trees, keep some trees.

Unit 2 Forces and movement

Topic 2.1 Forces around us

Focus

- 1 Answers can be written or drawn.

Things I push
skateboard wheelbarrow ball
Things I pull
comb kite toy boat

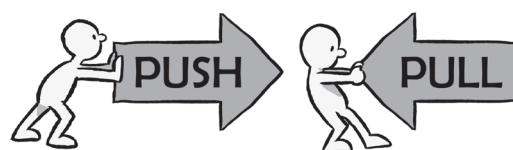
Practice

- 2 a The elephant is using a **big force**.
b The ant can push a leaf with a **small force**.
c I can move scissors with a **small force**.
d The train needs a **big force** to make it move.

Challenge

- 3 a A push is a force **away from you**.
b A pull is a force **towards you**.
c Forces can make things **move/stop**.
d Forces can make things **move/stop**.

4



Topic 2.2 Changing shape

Focus

1

a	m	e	a	s	u	r	e	t
s	f	u	b	a	z	h	p	w
o	i	p	a	t	t	e	r	n
b	a	i	q	e	p	i	e	b
j	b	c	h	a	n	g	e	x
e	d	l	k	j	g	h	f	d
c	s	a	p	o	i	t	u	y
t	z	d	a	q	f	w	r	t
u	b	a	s	h	a	p	e	e

Practice

2

Number of rolls	Number of paperclips long
10	2
20	3
30	4
40	6

Challenge

- 3 a 7 paperclips
b 2 paperclips
c 5 paperclips
d The more you roll the clay, the **longer it gets**.
e Correct answers describe any of the following:
- more force used, for example, Max was pushing harder on the clay.
 - different sized pieces of clay, for example, Max had more clay.

- different clay used, for example, Mina's clay was harder.
- temperature of clay affected its softness, for example, Max's clay was warmer.

Topic 2.3 Changing speed

Focus

1				
	Big push	Small push	Small push	Big push
				
	Fast or slow?	Fast or slow?	Fast or slow?	Fast or slow?

Practice

- 2 Pull the brakes to **slow down**.
Push the pedals to **speed up**.

Challenge

- 3 a She can go faster by pushing harder.
b She can slow down by pushing less hard or by dragging her foot on the floor to make a force in the other direction.

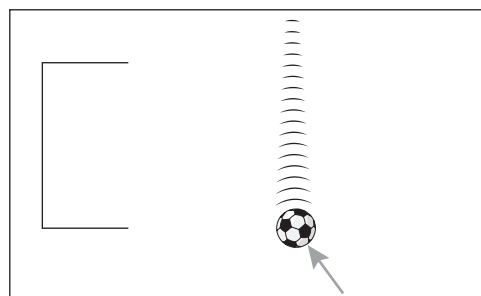
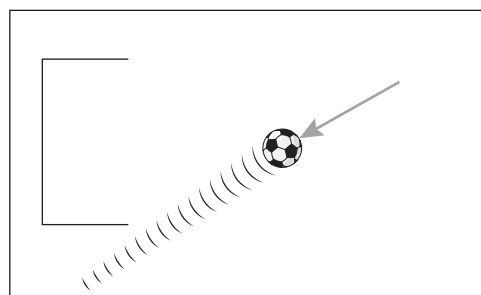
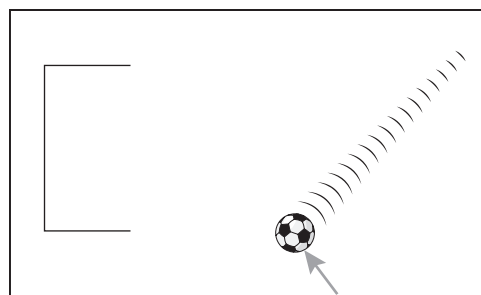
Topic 2.4 Changing direction

Focus

- 1 a A bigger push will make an object move **faster**.
b A smaller push will make an object move **slower**.
c Forces can make moving objects change **direction**.
d Pushes and pulls can make moving objects **turn**.

Practice

2



Learners should draw arrows in approximately the same directions as shown.

Challenge

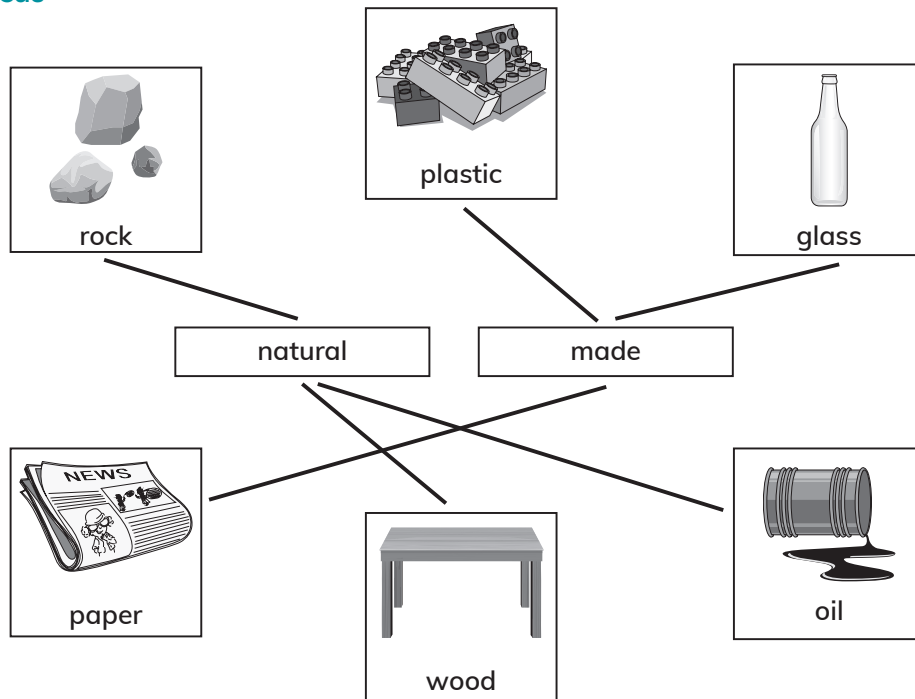
- 3 The predictions will depend on the learner.
4 Learner's own table entries.
5 A: can fall fast or slowly. B: falls slowly. C: falls fast.
Correct if prediction matches results.
6 A and B change **direction**.
C does not **change direction** until it hits the floor.

Unit 3 Getting materials right

Topic 3.1 Natural and made materials

Focus

1

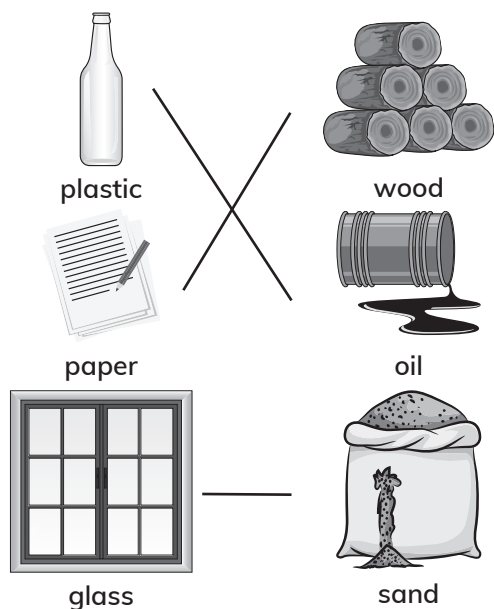


Practice

2

Made materials

Natural materials



Challenge

3

- a Plastic is made from **oil**.
- b Paper is made from **wood**.
- c Glass is made from **sand**.

4

- a Sheep
- b To make clothes/jumpers/gloves and scarves
- c Natural

Topic 3.2 Properties of materials

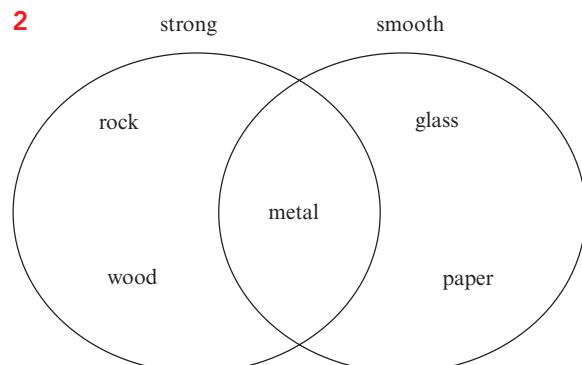
Focus

1

Object	Material	Characteristics
	plastic	rough, strong
	metal	smooth, flexible
	wood	strong, shiny
	glass	soft, flexible
	fabric	waterproof, smooth

Practice

2



Challenge

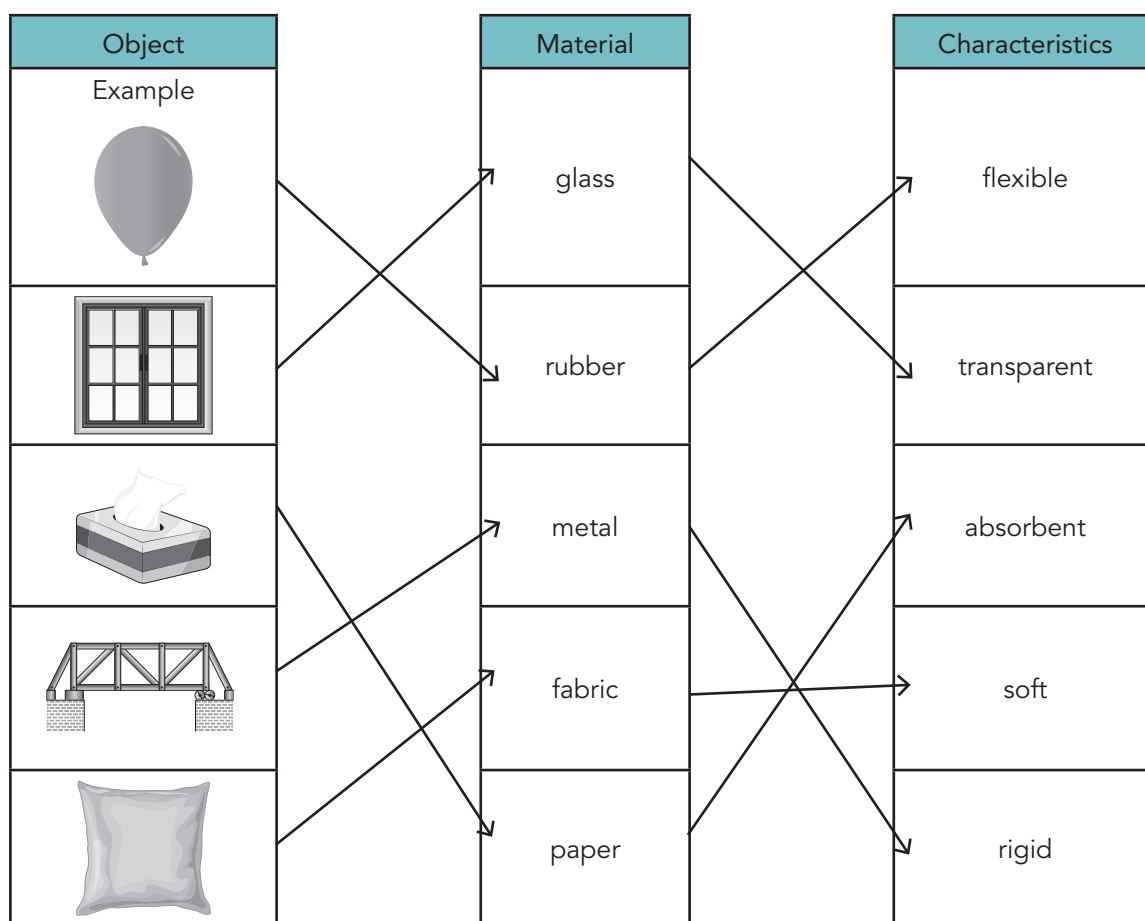
3

- a Most plastic is **waterproof** and **strong**.
- b All glass is **hard** and **rigid**.
- c Most rock is **strong** and **rigid**.
- d Most fabric is **absorbent** and **flexible**.

Topic 3.3 Using the right material

Focus

1



Practice

- 2 a A chair can be made of wood because wood is **strong**.
- b A window can be made of glass because glass is **transparent**.
- c A bottle can be made of plastic because plastic is **waterproof**.
- d A T-shirt can be made of fabric because fabric is **soft**.

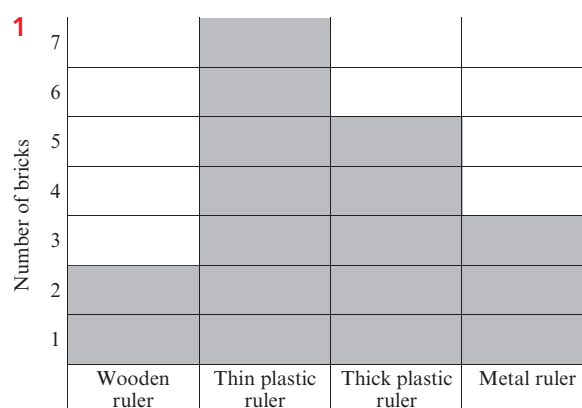
Challenge

- 3 a The windows are glass because glass is **transparent**.
- b The seats are fabric because fabric is **soft / flexible**.
- c The body is metal because metal is **strong**.
- d The tyres are rubber because rubber is **flexible**.

Topic 3.4 Testing materials

Focus

1



Practice

- 2 a The thin plastic ruler
 b The wooden ruler
 c The thin plastic ruler
 d No
 e The wooden ruler is less flexible than the metal ruler.

Challenge

- 3 a 5
 b 2
 c Correct answers show an understanding that the shape of the rulers affected their flexibility. For example, one was thick and the other was thin.
 d Correct answers will name a material that would not make a useable ruler, such as rubber, fabric or glass.
 e Look for an explanation of why the material given in part d would make an unusable ruler. Rubber and fabric are flexible; glass is weak, for example.

Topic 3.5 Changing materials

Focus

1



Practice

- 2 a Solid
 b Solid
 c Solid
 d Liquid
 e Liquid

Challenge

- 3 a A bar of chocolate is a **solid**. If you make a bar of chocolate hot it will **melt**. This makes the chocolate into a **liquid**. If you let the chocolate get cold again it will turn back into a **solid**.
 b Butter, sugar and eggs are used to make a cake. These three **different** materials are made into a **mixture**. They are then **heated** in an oven to make the cake.

Unit 4 Humans and animals grow

Topic 4.1 Comparing animals

Focus

1

Numbers of animals	5				
	4				
	3				
	2				
	1				
		bird	spider	cat	ant
					

Practice

2

0 legs	2 legs	4 legs	6 legs	8 legs

3

Numbers of animals	5					
	4					
	3					
	2					
	1					
		0 legs	2 legs	4 legs	6 legs	8 legs

Challenge

- 4**
- a** 2
 - b** 4
 - c** 9
 - d** 1

Topic 4.2 Growing

Focus

- 1**
- a** The baby monkey will grow taller. ☒
 - b** The baby monkey will grow more arms. ☒
 - c** The baby monkey will grow longer fur. ☒
 - d** The baby monkey will grow feathers. ☒
 - e** The baby monkey will grow a longer tail. ☒

Practice

- 2**
- a** Changes
 - b** Stays the same
 - c** Changes
 - d** Stays the same
 - e** Changes

Challenge

- 3**
- a** 2
 - b** 4
 - c** 3
 - d** The cat is fully grown/an adult after 10 months.

Topic 4.3 Inheriting characteristics

Focus

- 1**
- a** Identical/inherit
 - b** Inherit/identical
 - c** Fingerprint
 - d** Characteristic

Practice

- 2**
- a** Dad
 - b** Mum
 - c** Mum
 - d** Mum
 - e** Dad
 - f** Dad

Challenge

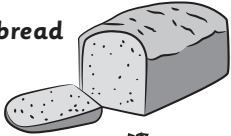
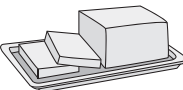
3

Fingerprint	Loop	Whorl	Arch
Names	Zara	Sofia	Arun
	Marcus	Leah	

Topic 4.4 Keeping healthy

Focus

1

Eat lots	Eat some	Eat a little
bread  carrot  pineapple  noodles 	milk  cheese  meat 	chocolate 

Practice

- 2 Answers will depend on learners' choices of food. The meal should be food that appears in the 'Eat lots' and 'Eat some' section of the food triangle. Mark as incorrect if food from the 'Eat a little' section is used.

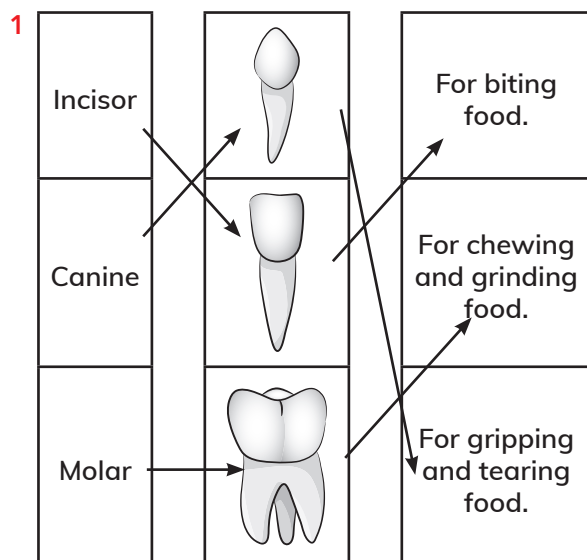
Challenge

3

	^a w	a	^b s	h				
			w					^c g
	^d e	x	e	r	c	^e i	s	e
			e			l		r
^f d	i	e	t			l		m
						n		s
						e		
						s		
						s		

Topic 4.5 Teeth

Focus



Practice

- 2
- Only eat a little food that has lots of **sugar**.
 - Make your teeth strong by eating food that has **calcium**.
 - Brush your teeth twice a day with **toothpaste**.
 - Use toothpaste with **fluoride**.
 - Go to the **dentist**.

Challenge

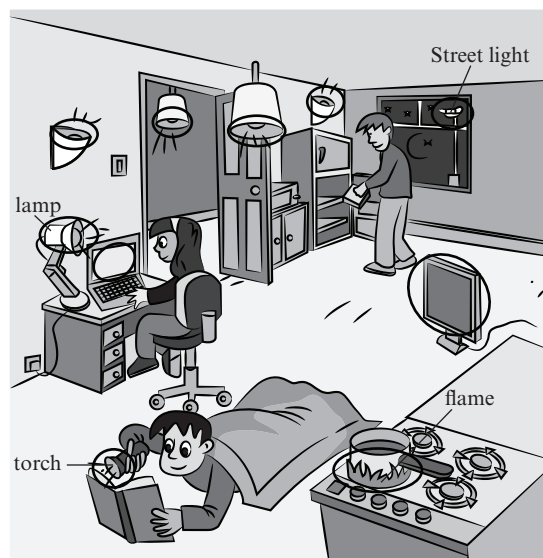
- 3
- Molar
 - Incisor
 - Canine
- 4 The root

Unit 5 Light

Topic 5.1 Light sources

Focus

1 and 2



- 3 A drawing of a flame, computer screen, star, wall lamp, street light or desk lamp.

Practice

Yes (✓): flashlight, lamp, match, Sun
No (✗): doorbell, clouds

Challenge

- 5
- Sophia and Zara
 - Arun
 - Sofia is **right** because she knows the Sun **makes its own light**.
- 6 Arun is **wrong** because the Moon **does not make its own light**.

Topic 5.2 Darkness

Focus

- 1
- White plate – easy
 - Dark cola – hard
 - Silver can – easy
 - Dark shoe – hard

Practice

- 2 First: It is too dark to see.
 Second: There is some light it is hard to see.
 Third: It is light, she can see well.

Challenge

- 3 a So that she can see things/so that she can be seen
 b So that other people can see her
 c Because the screen is bright/lit
 d The car has lights/ she has a head lamp/ reflective jacket
 e A flashlight/a white object

Topic 5.3 The Sun appears to move!

Focus

- 1 Sun drawn in the middle of the sky, towards the top.

Practice

- 2 10 am – low – to one side, noon – high in the middle. 4pm – low on the other side

Challenge

- 3 An arrow starting low on one side, arching high up and then return low on the other side.

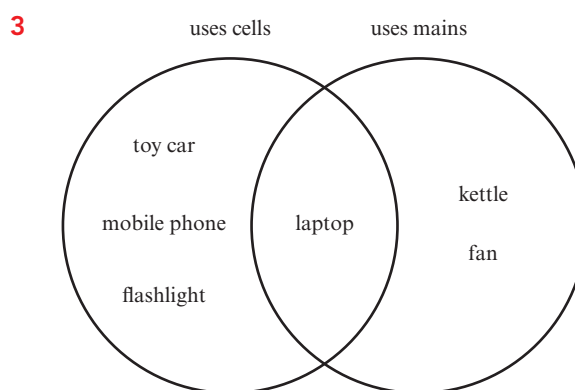
Unit 6 Electricity

Topic 6.1 Where do we use electricity?

Focus

- 1 Hair dryer – yes
 Television – yes
 Laptop – yes
 Plant – no
 Mobile phone – yes
 Radio – yes
 Flashlight – yes
- 2 Drawings of any two electrical appliances

Practice



- 4 a It needs electricity/we carry it around.
 b It needs a lot of electricity/we don't carry it around.

Challenge

- 5 a Wool
 c Cotton and wood
 d Hair
 e Coconut string
 f They are safe, bright and can be switched on and off quickly.

Topic 6.2 Keep safe with electricity

Focus



- 2 a Keep water away from **electricity**.
 b Do not touch damaged **wires**.
 c Never play with electricity, it can kill **you**.

Practice

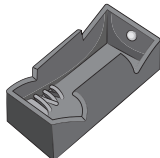
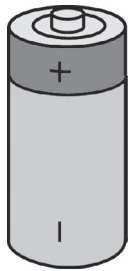
- 3** Picture 1 – If you stick things into a wall socket you might be electrocuted, hurt or killed (mains electricity is very dangerous).
 Picture 2 – Water is dangerous near electricity or you might be electrocuted, hurt or killed (the mains electricity is very dangerous).
 Picture 3 – Water is dangerous near electricity or you might be electrocuted, hurt or killed (the mains electricity is very dangerous).
 Picture 4 – Mains cables carry mains electricity which is dangerous or you might be electrocuted, hurt or killed (mains electricity is very dangerous).
- 4** Two rules similar to these:
 Never touch a damaged wire.
 Never touch a wall socket.
 Water is dangerous near electricity.

Challenge

- 5 a** Children should not touch mains wall sockets as they carry powerful or dangerous mains electricity. Or mains electricity can hurt or kill you. Or you might be electrocuted.
- b** A broken mains appliance may have wires which you could touch and get electrocuted. Or the electricity could hurt or kill you.
- c** Any answer which gives sensible advice about electrical safety.
- 6** A drawing might include: water near electricity, a damaged cable, wet hands near electricity, a child near mains electricity, a trailing lead on the floor or any other clear danger.

Topic 6.3 Making circuits

Focus

Name	Object	What it does
lamp		Makes light
cell holder		Holds a cell
lamp holder		Holds a lamp
cell		Stores electricity

- 2** Any reasonable drawn and complete circuit

Practice

- 3** Circuit 1 – the lamp does light up.
 Circuit 2 – the lamp does not light up.
 Circuit 3 – the lamp does not light up.
 Circuit 4 – the lamp does light up.
- 4** Any incomplete circuit with an explanation that it is not complete or that a wire or component is missing or not connected.

Challenge

- 5 a** Arrows drawn clockwise or anticlockwise on each picture showing a flow, one way around the circuit.
- b** C
- c** Because the three cells give more electricity/energy/push than one or two cells.
- d** Circuit C with any wire or component missing or disconnected.
- e** It may be brighter, or the lamp might 'blow' or break.